

ANTON KUEHR

2628 PZ Delft, Netherlands
anton.kuehr@web.de
+49 (0) 157 74441546
linkedin.com/in/anton-kuehr/
antonkuehr.github.io/akrobotics/

HARD SKILLS

- Programming in Python & C++
- Robot Operating System (ROS 2)
- Linux system administration
- PyTorch, OpenCV & YOLO
- GitLab & Docker
- Electronics & soldering
- Fusion 360 (design, FEM)
- FDM & resin 3D printing
- MATLAB (control strategies)
- MS Office applications
- Adobe Photoshop & Premiere
- Agile project management (Scrum)

LANGUAGES

- German (native)
- English C2/TOEFL 115 (proficient)
- French B1+ (intermediate)
- Swedish and Dutch (beginner)

INTERESTS

- Beach volleyball, table tennis, Swimming, ice-skating, skiing
- Favourite books: Thinking fast and slow, Factfulness, Sapiens
- Pop-Jazz improvisation on the piano
- Favorite YT-channels: Veritasium, Tom Scott, Practical Engineering

EDUCATION

09/2026 - 08/2028 **Delft University of Technology**
MSc in Robotics - Focus on dynamics and control, machine learning, perception, planning, robot software, human–robot interaction.
Robotics Student Association: Humanoid Project Echo – Skin Team Lead

10/2022 - 04/2026 **Cologne University of Applied Sciences**
BSc ME – Product Engineering and Context - Final grade: 1.1 (excellent).
Project-based programme combining mechanical engineering, product development, robotics, and interdisciplinary engineering design.

04/2023 - 03/2026 **Deutschlandstipendium**
Germany Scholarship: national merit-based public–private scholarship, co-funded by the German Federal Government and Alfred Schütte GmbH.

09/2013 - 07/2021 **Gymnasium Kreuzgasse, Cologne**
General qualification for university entrance, final grade: 1.3 (excellent), emphasis on sciences and languages (French-German bilingual school)

JOBS

04/2025 - 04/2026 **Prototyping and Concepting Intern**
Furhat Robotics: exploring new product concepts, improving existing designs, and developing alternative solutions to hardware and software challenges for the Furhat robot

03/2024 - 03/2025 **Student Tutor & Scrum Master**
CUAS: Acting as Scrum Master, coaching engineering courses & projects

11/2023 - 06/2024 **Sales Development Representative**
Perfect-Id GmbH, data solutions: customer acquisition, planning and process optimisation

08/2021 - 01/2022 **Food Delivery by Bike**
Wolt Enterprises Germany, Cologne

VOLUNTEER

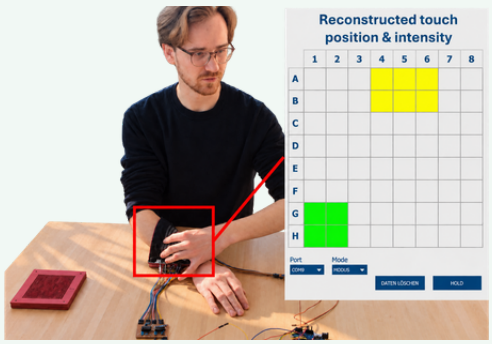
11/2025 - 02/2026 **Erasmus Ambassador**
Guidance and support for subsequent exchange students

10/2022 - 04/2026 **Student Council Member**
Advocating student concerns and sustainable initiatives on campus

07/2024 - 09/2024 **Jury Member and Workshop Leader**
Contributing to the recognition of innovative teaching formats, fostering dialogue between students and staff at the “Day for Excellent Teaching”

04/2022 - 06/2022 **Relief Worker**
Supporting victims of the Ahr Valley floods in July 2021 by joining the DZN aid organisation, contributing to reconstruction efforts, recruiting new volunteers, and providing humanitarian assistance

PORTFOLIO - SELECTED ROBOTICS PROJECTS

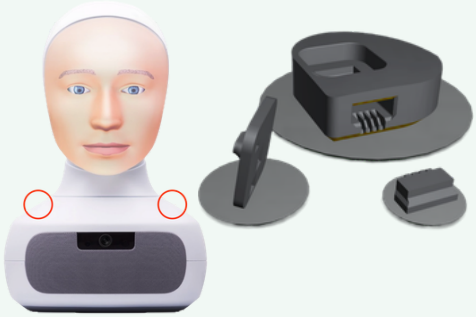


Left: Me applying pressure to the prototype
Right: Reconstructed touch position and intensity

09/2024 - Present **Robotic Skin Development**

BSc Project → RSA Project Echo, TU Delft

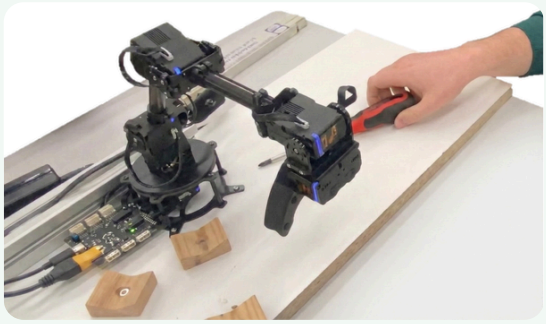
- As BSc project built a tactile skin prototype using conductive thread to localise pressure on robotic surfaces.
- Now leading the skin team of RSA Project Echo, building a humanoid aimed at competing in RoboCup Rescue 2028.
- Researching scalable sensing and soft-skin concepts for touch localisation, approximate force sensing, and protection.



Left: Furhat Robot, microphone position in red
Right: Redesigned microphone implementation in slicer

10/2025 - 04/2026 **Improving Furhat's Audio System**

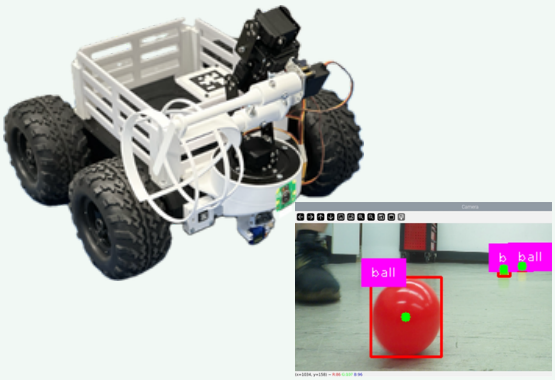
- Redesigned Furhat's MEMS microphone implementation, optimising mounting, isolation, and acoustic channel geometry through CAD and rapid prototyping.
- Built an automated ASR test setup to compare microphone systems across multiple SNR conditions.
- Reduced mean word error rate by 9.3%, with statistically significant improvements across all tested SNRs.



Collaborative pick-and-place setup

04/2024 - 03/2025 **Designing and Implementing a Cobot**

- Developed a voice-controlled cobot that interprets spoken requests, detects the requested tool and fetches or stows it.
- Built a ROS 2 architecture integrating Whisper-based speech processing, YOLO11-OBB object detection, and robot-arm control.
- Derived closed-form inverse kinematics for the 4-DOF arm to achieve reliable grasping poses despite limited degrees of freedom.



Left: Modified autonomous rover
Right: Custom YOLO object-detection output

09/2023 - 08/2024 **Developing an Agricultural Robot**

- Developing an agricultural robot that can detect, approach, pick and store apples/plastic balls autonomously.
- Programming the dynamics of a rover and robotic arm on Arduino and Raspberry Pi using inverse kinematics and PID control.
- Implementing computer vision based on YOLO models in native PyTorch format with OpenCV image processing.
- Comparing CV with a time-of-flight-based scanning approach.
- Designing and testing (FEM) components in Fusion 360.
- Using FDM and resin 3D printing for rapid prototyping.